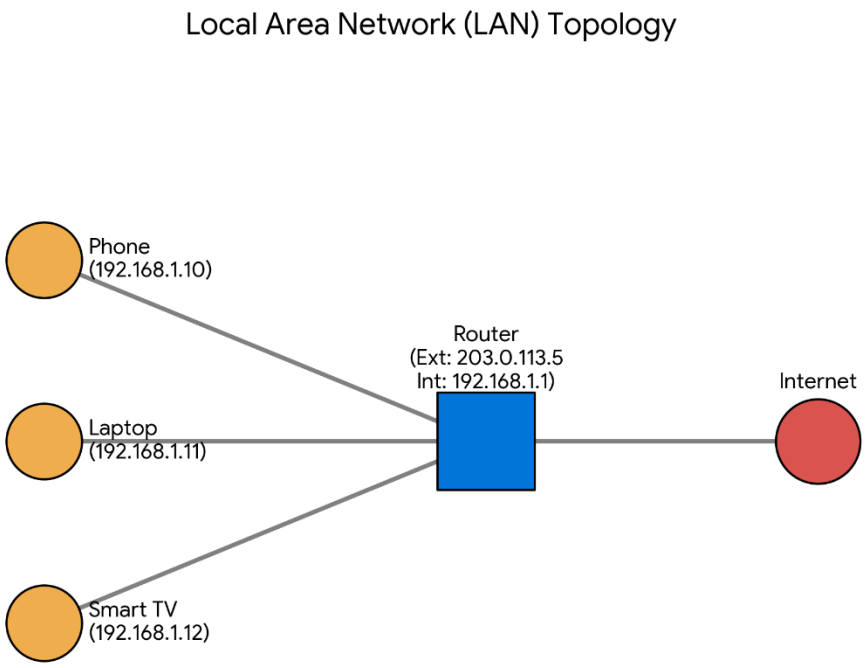
IP routing and routing protocols:-

Tasks:-

1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

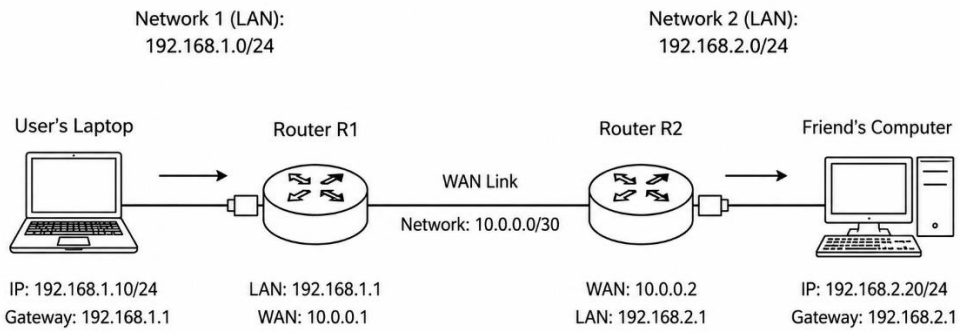
Ans. :-



**2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.**

Ans. :-



Step 1 — My laptop builds the packet:-

My laptop wants to send data to 192.168.2.20. It checks its own subnet mask (/24) and realizes 192.168.2.20 is not on its local network (192.168.1.0/24). Since the destination is remote, the OS doesn't try to find it directly — it sends the packet to its default gateway, 192.168.1.1 (Router A).

Step 2 — Router A receives the packet and checks its routing table:-

The router does a longest prefix match — it finds 192.168.2.0/24 matches directly, more specific than the default route, so it forwards out its WAN interface toward 10.0.0.2 (Router B).

Step 3 — Packet crosses the WAN link:-

Router A re-frames the packet with new Layer 2 (MAC) addressing for the WAN link, but the IP header is untouched: still source 192.168.1.10, destination 192.168.2.20. This is the key routing principle — routers change the "envelope" (MAC frame) at every hop, but never the IP addresses inside, unless NAT is involved.

Step 4 — Router B receives the packet :-

Router B sees 192.168.2.20 falls inside 192.168.2.0/24, which is directly connected to its LAN interface. No further hop needed — it ARPs for the friend's computer's MAC address and delivers the frame directly.

Step 5 — Friend's computer receives the packet:-

The friend's machine sees the packet addressed to its own IP, processes it, and (if replying) reverses the exact same process — checking its own routing table, sending to its default gateway (192.168.2.1), and the packet retraces the path back through Router B → Router A → my laptop.

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Ans. :-

Routing Table – Head Office Router (R1):-

Destination Network	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	10.0.12.2	Serial0/0/0
192.168.20.0	255.255.255.0	10.0.13.2	Serial0/0/1
192.168.30.0	255.255.255.0	10.0.14.2	Serial0/1/0

Routing Table – Ahmedabad Branch Router (R2) :-

Destination Network	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Directly Connected	GigabitEthernet0/0
192.168.20.0	255.255.255.0	10.0.12.1	Serial0/0/0
192.168.30.0	255.255.255.0	10.0.12.1	Serial0/0/0

Routing Table – Surat Branch Router (R3) :-

Destination Network	Subnet Mask	Next Hop	Interface
192.168.20.0	255.255.255.0	Directly Connected	GigabitEthernet0/0
192.168.10.0	255.255.255.0	10.0.13.1	Serial0/0/0
192.168.30.0	255.255.255.0	10.0.13.1	Serial0/0/0

Routing Table – Vadodara Branch Router (R4) :-

Destination Network	Subnet Mask	Next Hop	Interface
192.168.30.0	255.255.255.0	Directly Connected	GigabitEthernet0/0
192.168.10.0	255.255.255.0	10.0.14.1	Serial0/0/0
192.168.20.0	255.255.255.0	10.0.14.1	Serial0/0/0

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Ans.:-

Comparison of Static Routing and Dynamic Routing (RIP/OSPF) :-

Feature	Static Routing	Dynamic Routing (RIP/OSPF)
Definition	Routes are manually configured by the network administrator.	Routers automatically learn and update routes using routing protocols like RIP or OSPF.
Configuration	Manual	Automatic
Best For	Small networks	Medium and large networks
Updates	Manual changes required	Automatically updates when the network changes

Static Routing :-

Advantages :-

1. **Simple to configure :-** Easy to set up in a small network with only a few routers.
   Exa. A small restaurant with one internet router and a few billing computers.
2. **Low resource usage :-** Uses very little CPU, memory, and bandwidth because no routing updates are exchanged.
   Exa. A home Wi-Fi network with a laptop, phone, and smart TV.
3. **More secure :-** Routes cannot be changed automatically, reducing the risk of malicious routing updates.
   Exa. A bank ATM network where routes rarely change.

Disadvantages :-

1. **Time-consuming :-** Every route must be entered manually.
   Exa. A company with 50 branch offices would require many manual routes.
2. **Doesn't adapt to failures :-** If a link fails, traffic stops until an administrator updates the route.
   Exa. An online shopping warehouse loses connection because of a cable failure.
3. **Difficult to manage:-** As the network grows, maintaining routes becomes complex.
   Exa. A nationwide retail chain with hundreds of stores.

Dynamic Routing :-

Advantages:-

1. **Automatically updates routes:-** Routers detect network changes and choose new paths automatically.
   Exa. Zomato or Swiggy data centers continue communicating even if one network link fails.
2. **Scales well :-** Suitable for large networks with many routers.
   Exa. Google, Amazon, and Microsoft Azure networks.
3. **Chooses the best path** :- OSPF selects the fastest or lowest-cost route, improving performance.
   Exa. A company's headquarters communicates with branch offices using the shortest available route.

Disadvantages :-

1. **More complex** :- Requires knowledge of routing protocols such as RIP or OSPF.
   Exa. Enterprise networks need trained network engineers.
2. **Uses more resources :-** Routing updates consume CPU, memory, and bandwidth.
   Exa. A small office with only two routers may not benefit from dynamic routing.
3. **Slower initial setup**:- Routers need time to exchange routing information and build routing tables.
   Exa. After a network restart, routers take a short time before all routes are available.

**5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.**

Ans.:-

Router R1 Configuration:-

Router> enable

Router# configure terminal

Router(config)# router ospf 1

Router(config-router)# router-id 1.1.1.1

Router(config-router)# network 192.168.1.0 0.0.0.255 area 0

Router(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router(config-router)# exit

Router(config)# end

Router# write memory

Router R2 Configuration:-

Router> enable

Router# configure terminal

Router(config)# router ospf 1

Router(config-router)# router-id 2.2.2.2

Router(config-router)# network 192.168.2.0 0.0.0.255 area 0

Router(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router(config-router)# exit

Router(config)# end

Router# write memory

Configuration Line	**Explanation**
enable	Enters privileged EXEC mode to access advanced commands.
configure terminal	Enters global configuration mode to configure the router.
router ospf 1	Starts the OSPF routing process with process ID **1**.
router-id 1.1.1.1	Assigns a unique Router ID to Router R1.
router-id 2.2.2.2	Assigns a unique Router ID to Router R2.
network 192.168.1.0 0.0.0.255 area 0	Advertises the **192.168.1.0/24** network into OSPF Area 0.
network 192.168.2.0 0.0.0.255 area 0	Advertises the **192.168.2.0/24** network into OSPF Area 0.

Configuration Line	Explanation
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link between the two routers (10.0.0.0/30).
exit	Exits OSPF configuration mode.
end	Returns to privileged EXEC mode.
write memory	Saves the running configuration to the startup configuration so it remains after a reboot.